In [1]:

```
import numpy as np
```

In [2]:

```
arr = np.array([
    [1,2,3,4,5],
    [5,6,7,8,9],
    [1,3,5,7,9],
    [2,4,6,8,0],
    [10,11,12,13,14]
])
arr
```

Out[2]:

```
array([[ 1,  2,  3,  4,  5],
       [ 5,  6,  7,  8,  9],
       [ 1,  3,  5,  7,  9],
       [ 2,  4,  6,  8,  0],
       [10, 11, 12, 13, 14]])
```

Slicing the Array

array_name[index_of_subarray, index_of_elements

In [4]:

```
arr[3,4]
```

Out[4]:

```
0
```

In [6]:

```
arr[-1,-3]
```

Out[6]:

```
12
```

In [7]:

```
arr[1:3, 1:4]
```

Out[7]:

```
array([[6, 7, 8],
       [3, 5, 7]])
```

In [8]:

```
arr[:,:]
```

Out[8]:

```
array([[ 1,  2,  3,  4,  5],
       [ 5,  6,  7,  8,  9],
       [ 1,  3,  5,  7,  9],
       [ 2,  4,  6,  8,  0],
       [10, 11, 12, 13, 14]])
```

In [9]:

```
arr[::-1, ::]
```

Out[9]:

```
array([[10, 11, 12, 13, 14],
       [ 2,  4,  6,  8,  0],
       [ 1,  3,  5,  7,  9],
       [ 5,  6,  7,  8,  9],
       [ 1,  2,  3,  4,  5]])
```

In [10]:

```
arr[::-1, ::-1]
```

Out[10]:

```
array([[14, 13, 12, 11, 10],
       [ 0,  8,  6,  4,  2],
       [ 9,  7,  5,  3,  1],
       [ 9,  8,  7,  6,  5],
       [ 5,  4,  3,  2,  1]])
```

In [11]:

```
arr[0:5:2, 0:5:3]
```

Out[11]:

```
array([[ 1,  4],
       [ 1,  7],
       [10, 13]])
```

Joining the array

In [14]:

```
arr1 = np.array([1,2,3,4],dtype='int32')
arr2 = np.array([5,6,7,8],dtype='int32')
```

In [17]:

```
arr3 = np.concatenate((arr1, arr2))
arr3
```

Out[17]:

```
array([1, 2, 3, 4, 5, 6, 7, 8])
```

Joining 2-D Array

In [19]:

```
arr11 = np.array([[1,2,3,4],[2,3,4,5]])
arr12 = np.array([[4,5,6,7],[5,6,7,8]])
# Joining array on rows
arr13 = np.concatenate([arr11, arr12])
arr13
```

Out[19]:

```
array([[1, 2, 3, 4],
       [2, 3, 4, 5],
       [4, 5, 6, 7],
       [5, 6, 7, 8]])
```

In [20]:

```
# Joining array over columns
arr14 = np.concatenate([arr11, arr12], axis=1)
arr14
```

Out[20]:

```
array([[1, 2, 3, 4, 4, 5, 6, 7],
       [2, 3, 4, 5, 5, 6, 7, 8]])
```

Splitting an Array

In [21]:

```
arr21 = np.array([1,2,3,4,5,6,7,8,8,9,8,5,3,2,5,6])
arr21
```

Out[21]:

```
array([1, 2, 3, 4, 5, 6, 7, 8, 8, 9, 8, 5, 3, 2, 5, 6])
```

In [22]:

```
arr22 = np.array_split(arr21, 4)
arr22
```

Out[22]:

```
[array([1, 2, 3, 4]),
 array([5, 6, 7, 8]),
 array([8, 9, 8, 5]),
 array([3, 2, 5, 6])]
```

In [23]:

```
arr23 = np.split(arr21, 4)
arr23
```

Out[23]:

```
[array([1, 2, 3, 4]),
 array([5, 6, 7, 8]),
 array([8, 9, 8, 5]),
 array([3, 2, 5, 6])]
```

In [24]:

```
arr24 = np.array_split(arr21, 5)
arr24
```

Out[24]:

```
[array([1, 2, 3, 4]),
 array([5, 6, 7]),
 array([8, 8, 9]),
 array([8, 5, 3]),
 array([2, 5, 6])]
```

In [26]:

```
# unlike array_split,
#split function can not split the main array
#into arrays of different shape

# arr25 = np.split(arr21, 5)
# arr25
```

Splitting 2-D Array

In [27]:

```
arr26 = np.array([
    [1,2,3,4,5],
    [5,6,7,8,9],
    [1,3,5,7,9],
    [2,4,6,8,0],
    [10,11,12,13,14]
])
arr26
```

Out[27]:

```
array([[ 1,  2,  3,  4,  5],
       [ 5,  6,  7,  8,  9],
       [ 1,  3,  5,  7,  9],
       [ 2,  4,  6,  8,  0],
       [10, 11, 12, 13, 14]])
```

In [30]:

```
# Splitting Array on rows
arr27 = np.array_split(arr26, 3)
for i in arr27:
    print('Array',i)
```

```
Array [[1 2 3 4 5]
 [5 6 7 8 9]]
Array [[1 3 5 7 9]
 [2 4 6 8 0]]
Array [[10 11 12 13 14]]
```

In [32]:

```
# splitting array on columns

arr28 = np.array_split(arr26, 3, axis=1)
for i in arr28:
    print('Array',i, i.shape)
```

```
Array [[ 1  2]
 [ 5  6]
 [ 1  3]
 [ 2  4]
 [10 11]] (5, 2)
Array [[ 3  4]
 [ 7  8]
 [ 5  7]
 [ 6  8]
 [12 13]] (5, 2)
Array [[ 5]
 [ 9]
 [ 9]
 [ 0]
 [14]] (5, 1)
```

Searching in Array

In [33]:

```
arr31 = np.array([1,2,3,4,5,6,7,42,25,3,25,3,5])
arr31
```

Out[33]:

```
array([ 1,  2,  3,  4,  5,  6,  7, 42, 25,  3, 25,  3,  5])
```

In [36]:

```
pos = np.where(arr31 == 4)
pos
```

Out[36]:

```
(array([3], dtype=int64),)
```

In [37]:

```
np.where(arr31 == 5)
```

Out[37]:

```
(array([ 4, 12], dtype=int64),)
```

In [38]:

```
# Returns index of all even numbers
np.where(arr31%2 == 0)
```

Out[38]:

```
(array([1, 3, 5, 7], dtype=int64),)
```

In [39]:

```
# Returns index of all odd numbers
np.where(arr31%2 != 0)
```

Out[39]:

```
(array([ 0,  2,  4,  6,  8,  9, 10, 11, 12], dtype=int64),)
```

Searching on 2-D Array

In [40]:

```
arr32 = np.array([
    [1,2,3,4,5],
    [5,6,7,8,9],
    [1,3,5,7,9],
    [2,4,6,8,0],
    [10,11,12,13,14]
])
arr32
```

Out[40]:

```
array([[ 1,  2,  3,  4,  5],
       [ 5,  6,  7,  8,  9],
       [ 1,  3,  5,  7,  9],
       [ 2,  4,  6,  8,  0],
       [10, 11, 12, 13, 14]])
```

In [41]:

```
np.where(arr32 == 12)
```

Out[41]:

```
(array([4], dtype=int64), array([2], dtype=int64))
```

In [42]:

```
np.where(arr32 == 9)
```

Out[42]:

```
(array([1, 2], dtype=int64), array([4, 4], dtype=int64))
```

Sorting an Array

In [43]:

```
arr41 = np.array([1,2,3,4,5,6,7,42,25,3,25,3,5])
arr41
```

Out[43]:

```
array([ 1,  2,  3,  4,  5,  6,  7, 42, 25,  3, 25,  3,  5])
```

In [44]:

```
arr42 = np.sort(arr41)
arr42
```

Out[44]:

```
array([ 1,  2,  3,  3,  3,  4,  5,  5,  6,  7, 25, 25, 42])
```

Sorting 2-D Array

In [45]:

```
arr43 = np.array([
    [1,2,3,4,5],
    [5,6,7,8,9],
    [1,3,5,7,9],
    [2,4,6,8,0],
    [10,11,12,13,14]
])
arr43
```

Out[45]:

```
array([[ 1,  2,  3,  4,  5],
       [ 5,  6,  7,  8,  9],
       [ 1,  3,  5,  7,  9],
       [ 2,  4,  6,  8,  0],
       [10, 11, 12, 13, 14]])
```

In [47]:

```
arr44 = np.sort(arr43)
arr44
```

Out[47]:

```
array([[ 1,  2,  3,  4,  5],
       [ 5,  6,  7,  8,  9],
       [ 1,  3,  5,  7,  9],
       [ 0,  2,  4,  6,  8],
       [10, 11, 12, 13, 14]])
```

In [ ]:

```
```